Marmara Üniversitesi
Mühendislik Fakültesi
Elektrik-Elektronik Mühendisliği

Laboratory Manual

EE 3011
Electronics I

Experiment (3)
Simulation of Diode Circuits using LTspice

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Simulation of Diode Circuits using LTspice

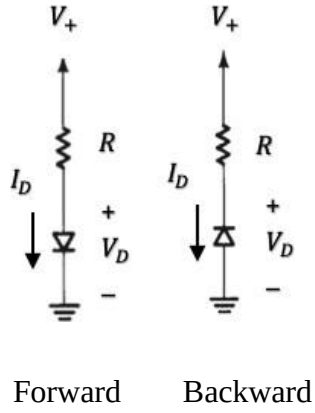
Objectives

- To study the IV characteristics of a standard diode (1N4001) by simulation.
- To simulate simple DC and AC diode circuits.

Pre-lab Work

Analyze the circuits below using the constant voltage diode model (V_{D0}=0.7 V).

Circuits



Forward Backward

Figure 1: Diode IV Circuit.

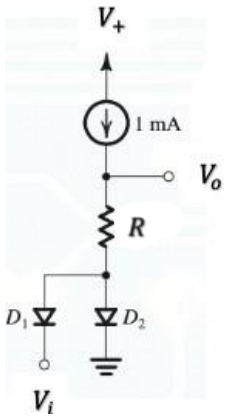


Figure 2: DC diode Circuit.

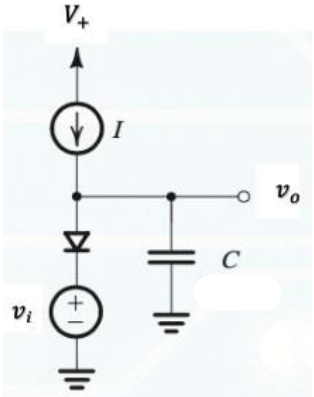


Figure 3: AC Diode Circuit

Procedure

Part (1): Diode IV Characteristics.

1. Connect the circuit in figure 1 (Forward). Use the circuit elements values given by the lab instructor. $V_{!}$ is a DC voltage source.
2. Use the DC Sweep function and simulate the response of the circuit to produce the IV curve of the diode (I_D vs. V_D). What is the $V_{D\#}$ of this diode? Record the results in table 1.
3. Repeat for the backward circuit. What is the $V_{\$\#}$ of this diode? Record the results in table 1.
4. Vary the value of R and repeat the procedure observing the effects (if any).

Part (2): DC Diode Circuit.

1. Connect the circuit in figure 2. Use the circuit elements values given by the lab instructor. $V_{!}$ and V_{i} are fixed DC voltage sources.
2. Simulate the circuit. Which diode is ON and which is OFF?

3. With V_i fixed, vary the value of R and repeat step 2 observing the effects on the diodes switching. Record your results in table 2.
4. With R fixed, vary V_i between the range -10 to +10 V and plot the output characteristics curve (V_o vs. V_i).

Part (3): AC Diode Circuit.

1. Connect the circuit in figure 3. Use the circuit elements values given by the lab instructor. I in the circuit is a DC current source and v_i is a sinusoidal signal with small amplitude (less than 10 mV) and a frequency f.
2. Simulate the circuit and plot the waveforms for v_i and v_o (versus time).
3. From the plot in step 2, find the phase shift between v_i and v_o. Record the result in table 3.
4. Vary I and observe the phase shift between v_i and v_o at each setting. Record your results in table 3. Find a value of I that gives a phase shift of ±45°?

Results and Analysis

Part (1): Diode IV Characteristics.

Table 1. Diode IV Curve.

R (kΩ)	Simulation V_{D0} (V)	Data sheet V_{D0} (V)	Simulation V_{Z0} (V)	Data sheet V_{Z0} (V)
1	0.7	0.72	- 0.7	- 0.72
10	0.7	0.72	- 0.7	- 0.72
100	0.7	0.72	- 0.7	- 0.72

1. Comment on the results in table 1.

The change in resistance values does not cause any change in voltage values.

2. Do the IV curves that you obtained resemble the standard diode characteristics? Explain.

Yes, the diode behaves like an open circuit up to a certain voltage value and does not pass current. Then, after a certain voltage value, it acts like a short circuit and starts to flow current.

Part (2): DC Diode Circuit.

Table 2. Simulated DC results.

R (kΩ)	Diode 1 (ON/OFF)	Diode 2 (ON/OFF)	V_0 (V)
1	OFF	ON	0.03
10	OFF	ON	0.041
100	OFF	ON	0.132
1k	OFF	ON	1

1. Comment on the results in table 2.

As the resistor value increases, the output voltage also increases.

2. Comment of the output characteristics curve.

The output voltage continues to increase exponentially as long as the resistance value continues to increase

Part (3): AC Diode Circuit.

Table 3. Simulated AC results.

I (mA)	Phase shift
1	180 µs
10	3.8 µs
100	1.3 µs
1000	0.1 µs
10000	0
100000	0

1. Which value of *I* gives a phase shift of ±45°?

It was not shown how to find phase shift in the lab and I cannot answer this question because I cannot find it on the internet.

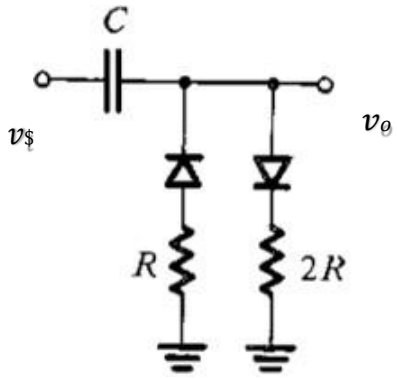
2. Explain the output waveform.

As the current value increases, the input and output graphs become more similar to each other and the phase shit between them decreases.

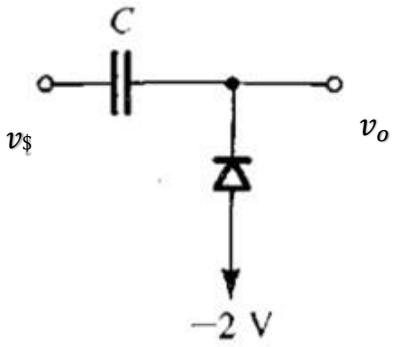
3. Comment on the circuit output characteristics curve.

As you increase the current value, the sine wave in the received sine signal becomes smoother and more distinct.

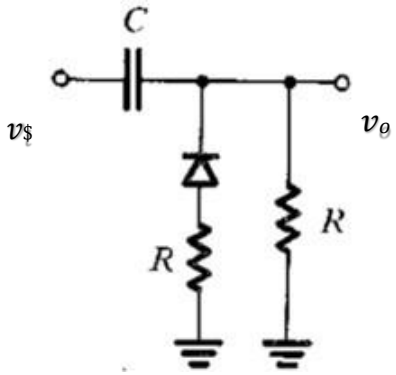
Lab Assignment



Circuit 1

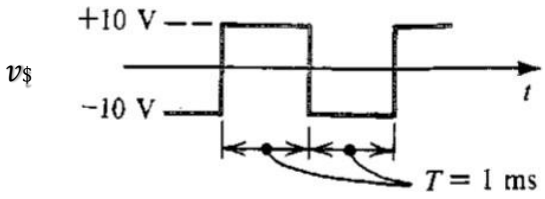


Circuit 2



Circuit 3

1. The lab instructor will assign one circuit to you with appropriate elements values. All diodes are 1N4001 model.
2. Connect the circuit using the circuit elements values given by the lab instructor. v_i is a periodic signal as shown below.
3. Simulate the circuit observing the waveform at the output (v_o).
4. By varying the amplitude of the input signal from 0 to 15 V, produce the transfer function curve of this circuit at the frequency given below.
5. Fix the amplitude at the value below but vary the signal period from 0.1 to 10 ms and make observations regarding the circuit operation.
6. Study the effect of the elements R and C (independently) on the operation of the circuit. Give conclusions.
7. What application can this circuit be good for? Explain.



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FAIRCHILD

November 2014

1N4001 - 1N4007
General-Purpose Rectifiers

Features

- Low Forward Voltage Drop
- High Surge Current Capability



DO-41
COLOR BAND DENOTES CATHODE

Ordering Information

Part Number	Top Mark	Package	Packing Method
1N4001	1N4001	DO-204AL (DO-41)	Tape and Reel
1N4002	1N4002	DO-204AL (DO-41)	Tape and Reel
1N4003	1N4003	DO-204AL (DO-41)	Tape and Reel
1N4004	1N4004	DO-204AL (DO-41)	Tape and Reel
1N4005	1N4005	DO-204AL (DO-41)	Tape and Reel
1N4006	1N4006	DO-204AL (DO-41)	Tape and Reel
1N4007	1N4007	DO-204AL (DO-41)	Tape and Reel

Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only. Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value							Unit
		1N 4001	1N 4002	1N 4003	1N 4004	1N 4005	1N 4006	1N 4007	
V_{RRM}	Peak Repetitive Reverse Voltage	50	100	200	400	600	800	1000	V
$I_{F(AV)}$	Average Rectified Forward Current .375 " Lead Length at T_A = 75°C	1.0							A
I_{FSM}	Non-Repetitive Peak Forward Surge Current 8.3 ms Single Half-Sine-Wave	30							A
I^2t	Rating for Fusing (t < 8.3 ms)	3.7							A^2sec
T_{STG}	Storage Temperature Range	-55 to +175							°C
T_J	Operating Junction Temperature	-55 to +175							°C

Thermal Characteristics

Values are at T_A = 25°C unless otherwise noted.

Symbol	Parameter	Value	Unit
P_D	Power Dissipation	3.0	W
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient	50	°C/W

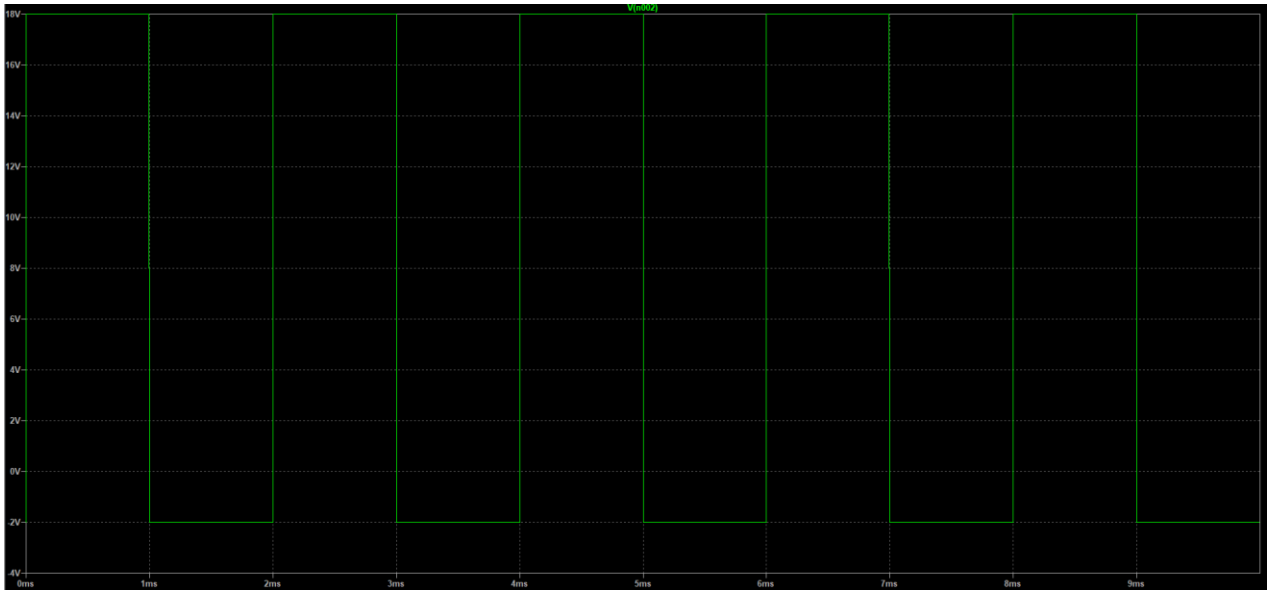
Electrical Characteristics

Values are at T_A = 25°C unless otherwise noted.

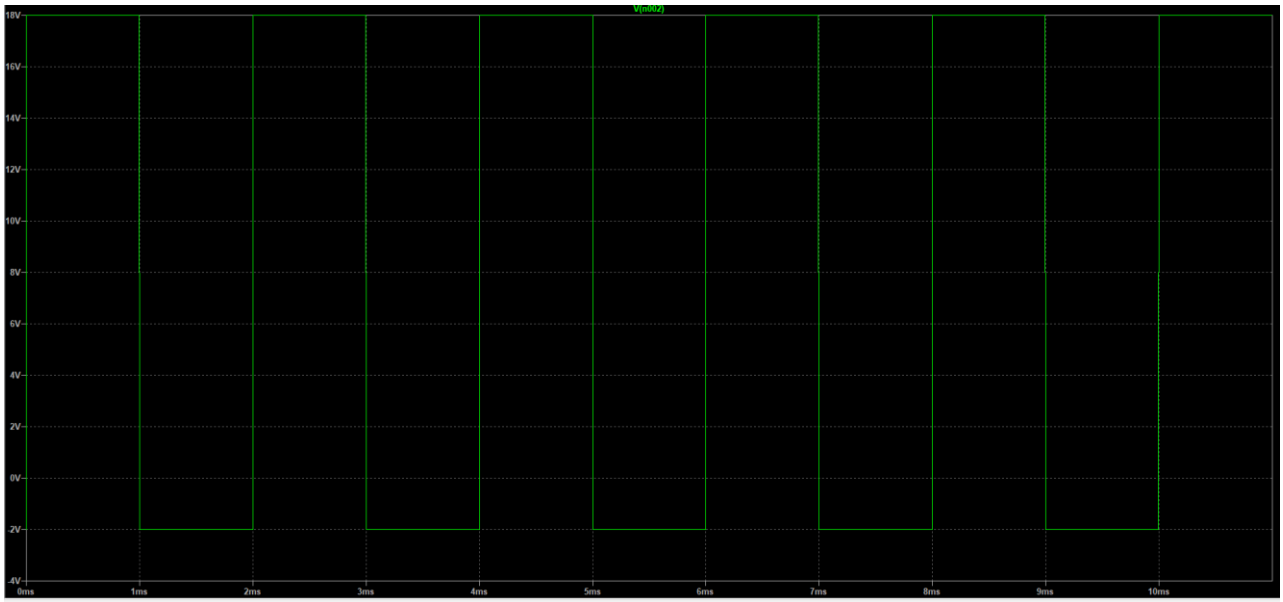
Symbol	Parameter	Conditions	Value	Unit
V_F	Forward Voltage	I_F = 1.0 A	1.1	V
I_{rr}	Maximum Full Load Reverse Current, Full Cycle	T_A = 75°C	30	µA
I_R	Reverse Current at Rated V_R	T_A = 25°C	5.0	µA
		T_A = 100°C	50	
C_T	Total Capacitance	V_R = 4.0 V, f = 1.0 MHz	15	pF

Lab Assigment:

Lab instructor chose second circuit.



I cannot change R values as there is no resistor in the circuit. I just change the C values of the capacitor



10 µF



100 μF

When I change the C values, there is no change in the graphics.